Abstraction: A Super Tool for Decisions under Uncertainty

One Mathematical Skeleton, Many Real-World Stories (C1–C4)

Contents

Evaluated criteria

C1 Identify the decision structure: alternatives, uncertain event, states of nature, consequences, payoffs, and units.

C2 Organise alternatives and states of nature in a complete payoff table.

C3 Calculate or construct the payoff for each alternative–state pair.

C4 Apply and interpret Maximax and Maximin decision criteria.

1 Abstraction: One Mathematical Skeleton, Many Real-World Stories

Problems:

1.1 A Surprising Connection

On Monday, Maya helps a café decide whether to install a small or large production capacity before customer demand is known. Across town, Leo advises a farmer who must plant crop A or crop B before the season turns dry or rainy. Meanwhile, Sara compares two investment projects before the market proves weak or strong.

At first, the problems seem unrelated: coffee cups, crops, and investments have different prices, risks, units, and consequences. Those are the story's *clothing*. Look beneath it, however, and the same mathematical skeleton appears:

$$\boxed{\text{choice before uncertainty}} + \boxed{\text{possible states}} + \boxed{\text{payoff for each combination}}.$$

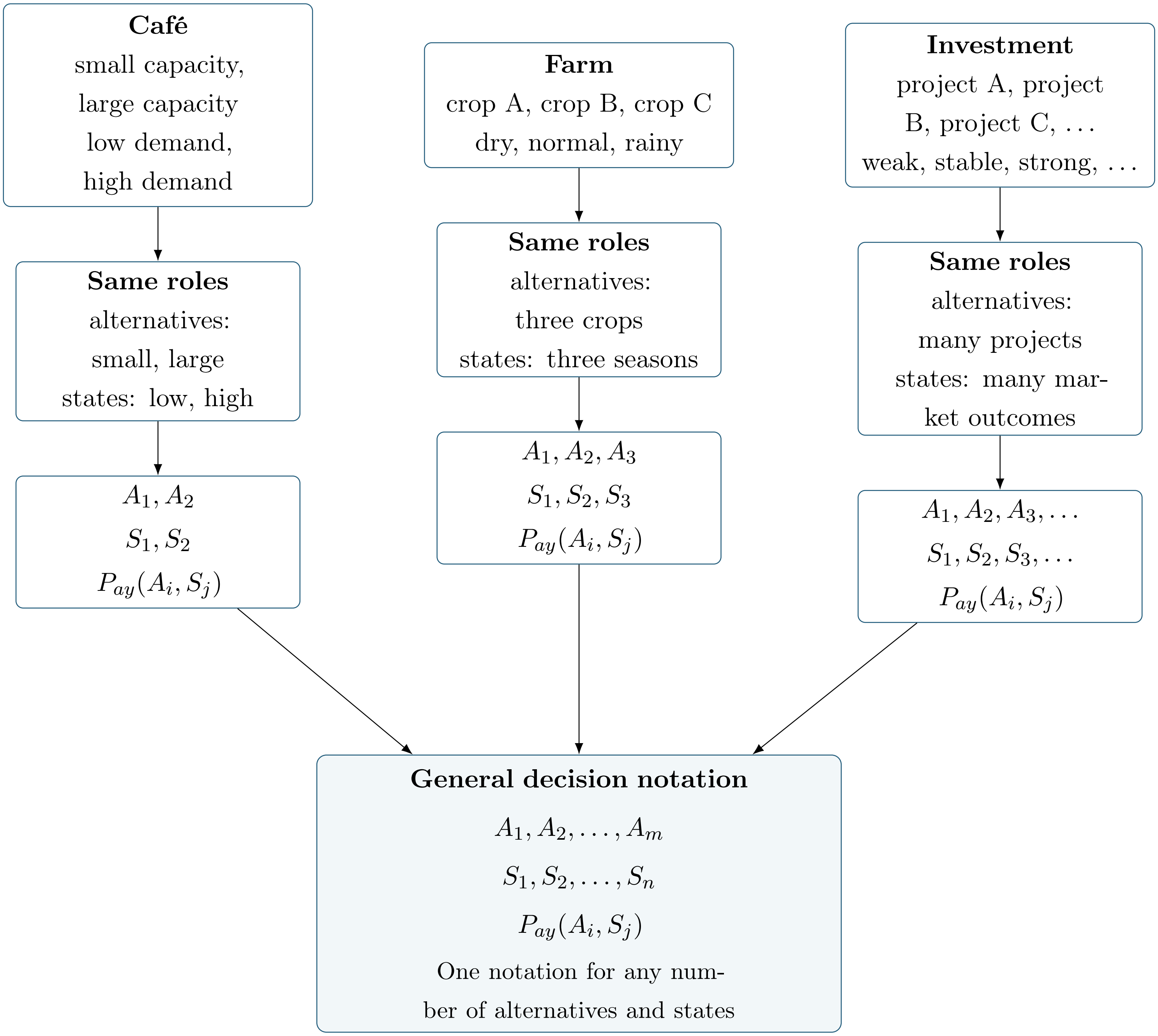
This pattern also appears when a company selects a production plan, a family evaluates a financial decision, a school enterprise plans output, or a government selects a policy. Recognising it is the super tool called **abstraction**.

1.2 From Different Stories to One Notation

The nouns change, but their mathematical *roles* stay the same. A decision-maker chooses an alternative A_i, an uncertain state S_j occurs, and the combination determines the payoff $P_{ay}(A_i, S_j)$.

For example, Maya's "large capacity", Leo's "crop B", and Sara's "project B" may all be labelled A_2. They are not the same real-world choice; they simply occupy the same role in their respective decision models.

The same abstraction works even when the number of alternatives or states changes:



The notation preserves the common structure without erasing the original context: choose one alternative A_i, observe one state S_j, and obtain the payoff $P_{ay}(A_i, S_j)$. The letters m and n allow the same notation to represent any number of alternatives and states.

1.3 The C1–C4 Abstraction Pipeline

The same seven-step method can be read in parallel with one small example. The left side records the procedure; the right side shows what that stage looks like mathematically.

1. Read the context
C1: identify

2. Extract alternatives, event, states, consequences, and units
C1: identify

Maya's café must choose between a small-capacity plan and a large-capacity plan before next month's customer demand is known. Demand may be low or high. Each capacity–demand combination produces a monthly profit, measured in thousand USD.

Element	Description	
Alternatives	**A1:** Small capacity A_1	**A2:** Large capacity A_2
Statesofnature	**S1:** Low demand S_1	**S2:** High demand S_2
Events	Customer demand after the capacity plan is selected	
Consequences	Monthly profit in thousand USD for each capacity–demand pair	

3. Build the payoff-table skeleton
C2: organise

4. Calculate or insert each payoff
C2–C3: construct

5. Check completeness and units
C2–C3: validate

Skeleton

Alternative	S_1	S_2
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Insert or construct each cell

$$\text{C2:} \qquad P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

$$\text{C3:} \qquad P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j),$$

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j),$$
$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j).$$

Completed and checked

Alternative	S_1	S_2
A_1	10	4
A_2	7	8

Every pair has one payoff; all payoffs use the same units.

6. Apply Maximax and/or Maximin
C4: compare

7. Interpret the result and limitations in context
C4: interpret

$$\boxed{A_{\text{Maximax}} = \arg\max_i \left\{ \max_j P_{ay}(A_i, S_j) \right\}}$$

$$\boxed{A_{\text{Maximin}} = \arg\max_i \left\{ \min_j P_{ay}(A_i, S_j) \right\}}$$

For the table above,

$$\begin{aligned} &\text{row maxima: } (10, 8) && \Rightarrow A_{\text{Maximax}} = A_1, \\ &\text{row minima: } (4, 7) && \Rightarrow A_{\text{Maximin}} = A_2. \end{aligned}$$

Decision statement. If Maya uses the optimistic Maximax criterion, she should choose small capacity (A_1) because its best possible profit is 10 thousand USD. If she prioritises protection against the worst modelled outcome, Maximin selects large capacity (A_2), whose worst payoff is 7 thousand USD. Neither rule uses probabilities, so the recommendation depends on the decision attitude and on the payoffs and states included in the model.

1.4 C1: Identify the Decision Structure

Start by asking: Who makes the decision? What is the economic or financial context? What can that person or organisation choose? What uncertain event occurs *after* the choice? Which states can it produce? What consequence belongs to each alternative–state pair? In what units will it be measured?

Write the m alternatives and n states as

$$A_1, A_2, \ldots, A_m \qquad \text{and} \qquad S_1, S_2, \ldots, S_n.$$

The order is important:

$$\text{Choose } A_i \quad \longrightarrow \quad \text{uncertainty is resolved as } S_j \quad \longrightarrow \quad \text{consequence and payoff } P_{ay}(A_i, S_j).$$

- The decision-maker controls the choice of A_i.
- The decision-maker does not control which S_j occurs.
- The pair (A_i, S_j) determines a consequence.
- The payoff assigns an economic or financial value to that consequence.

Once these elements have been identified, record them in a `\DecisionElementsTable`. The table separates the four pieces needed before constructing a payoff table:

$$\boxed{\text{alternatives}} \qquad \boxed{\text{states}} \qquad \boxed{\text{uncertain event}} \qquad \boxed{\text{consequence and units}}.$$

For example, suppose Maya's café must choose between a small-capacity plan and a large-capacity plan before next month's customer demand is known. Demand may be low or high. Each capacity–demand pair produces a monthly profit measured in thousand USD.

Element	Description	
Alternatives	**A1:** Small capacity A_1	**A2:** Large capacity A_2
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Consequences	Monthly profit in thousand USD for each capacity–demand pair	

The table makes the abstraction explicit:

$$A_i = \text{capacity choice}, \qquad S_j = \text{realised demand}, \qquad P_{ay}(A_i, S_j) = \text{monthly profit}.$$

A *consequence* may first be expressed in words. For example, "the large café has unused equipment when demand is low." Its numerical *payoff* might then be a profit of -2 thousand USD. The words describe what happens; the payoff gives that consequence a value that can later be placed in the payoff table. The units are part of that value.

1.5 C2: Give the Structure a Table

The payoff table organizes the decision structure identified in C1. **Alternatives go in rows**, **states go in columns**, and each cell contains the payoff associated with one alternative–state pair (A_i, S_j).

The general structure is:

Alternative	S_1	S_2	$\cdots$	S_n
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$\cdots$	$P_{ay}(A_1, S_n)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$\cdots$	$P_{ay}(A_2, S_n)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
A_m	$P_{ay}(A_m, S_1)$	$P_{ay}(A_m, S_2)$	$\cdots$	$P_{ay}(A_m, S_n)$

Thus,

$$P_{ay}(A_i, S_j)$$

is the payoff obtained when alternative A_i is chosen and state S_j occurs.

The table has m alternative rows and n state columns. Therefore, there are

$$mn$$

alternative–state pairs, and each pair must have exactly one payoff.

For a problem with two alternatives and two states, the payoff-table skeleton is:

Alternative	S_1	S_2
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Once the payoffs have been calculated or supplied, they replace the symbolic entries. For example:

Alternative	S_1	S_2
A_1	10	4
A_2	7	8

Every numerical entry must use the same payoff meaning and compatible units. The table is complete only when every pair (A_i, S_j) has exactly one corresponding value.

1.6 C3: Build the Value in Each Cell

In C2, each payoff-table cell was identified as

$$P_{ay}(A_i, S_j).$$

C3 determines the numerical value that goes into that cell.

For each pair (A_i, S_j), ask first:

Is the payoff given directly, or must it be calculated?

If the payoff is supplied directly, place that value in the corresponding cell. If it must be constructed from payoff components, use

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,[v(A_i) + v(S_j)] + f(A_i) + f(S_j).$$

Here,

- $q(A_i, S_j)$ is the quantity associated with the cell;
- $v(A_i)$ is a variable per-unit payoff component determined by the alternative;
- $v(S_j)$ is a variable per-unit payoff component determined by the state;
- $f(A_i)$ is a fixed payoff component determined by the alternative;
- $f(S_j)$ is a fixed payoff component determined by the state.

The result is one complete value for one payoff-table cell:

$$(A_i, S_j) \quad \longrightarrow \quad P_{ay}(A_i, S_j) \quad \longrightarrow \quad \text{insert in row } A_i,\text{ column } S_j.$$

When the payoff represents *profit*, construct revenue and cost separately:

$$P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

A general revenue model is

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j),$$

while a general cost model is

$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j).$$

Therefore, the workflow for a profit cell is

$$\boxed{\text{calculate } R(A_i, S_j)} \quad \longrightarrow \quad \boxed{\text{calculate } C(A_i, S_j)} \quad \longrightarrow \quad \boxed{P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)}.$$

For example, the cell in row A_1 and column S_2 is

$$P(A_1, S_2) = R(A_1, S_2) - C(A_1, S_2).$$

Once this value is calculated, it is placed directly in the corresponding position of the payoff table:

Alternative	S_1	S_2
A_1	$P(A_1, S_1)$	$\boxed{P(A_1, S_2)}$
A_2	$P(A_2, S_1)$	$P(A_2, S_2)$

Repeat the same process until every cell contains one payoff. When using these formulas:

- use only the components given in the context;
- multiply only per-unit values by quantity;
- keep fixed values separate from quantity;
- keep all values in compatible units;
- calculate revenue and cost separately before subtracting.

The objective of C3 is therefore not to choose an alternative. It is to complete the payoff table constructed in C2 by assigning exactly one correctly calculated payoff to every pair (A_i, S_j).

1.7 C4: View One Table Through Two Decision Lenses

C1–C3 construct a valid model and completed table. C4 asks how to compare its rows when no probabilities are supplied. The two criteria below answer different questions; neither is universally correct.

The Optimistic Lens: Maximax

Maximax selects the largest possible payoff. First define the row maximum for alternative A_i by

$$M_i = \max_{j \in \{1,\dots,n\}} P(A_i, S_j).$$

Then select the alternative (or alternatives) with the largest row maximum:

$$A_{\text{Maximax}} = \arg\max_{i \in \{1,\dots,m\}} \left\{ \max_{j \in \{1,\dots,n\}} P(A_i, S_j) \right\}.$$

The Protective Lens: Maximin

Maximin selects the largest worst-case payoff. First define the row minimum for alternative A_i by

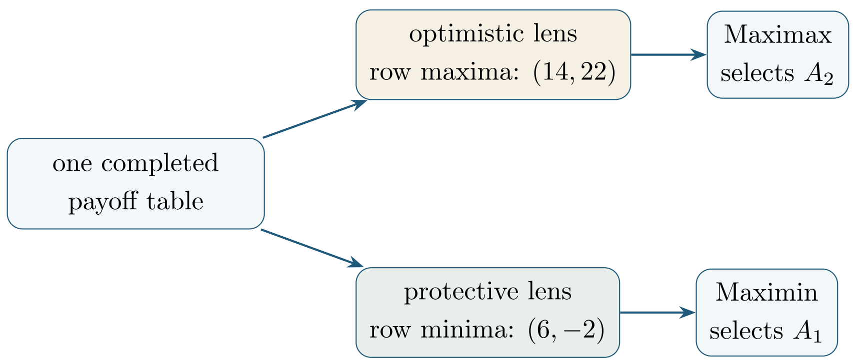
$$m_i = \min_{j \in \{1,\ldots,n\}} P(A_i, S_j).$$

Then select the alternative (or alternatives) with the largest row minimum:

$$A_{\text{Maximin}} = \arg\max_{i \in \{1,\ldots,m\}} \left\{ \min_{j \in \{1,\ldots,n\}} P(A_i, S_j) \right\}.$$

One small table makes the two lenses visible. Suppose Maya's café profits are in **thousand USD**:

Alternative	Low demand S_1	High demand S_2
Small capacity A_1	6	14
Large capacity A_2	-2	22



Thus, the optimistic lens favours large capacity's upside, while the protective lens favours small capacity's stronger worst case. If two alternatives share the relevant row maximum under Maximax or row minimum under Maximin, report a **tie**; do not choose one arbitrarily.

1.8 Zoom Back into the Real World

A calculation is not yet a complete recommendation. Maximax focuses on upside and may overlook severe downside. Maximin protects against the worst payoff represented in the model but may reject attractive opportunities. Neither rule uses probabilities. Maya should therefore say more than "A_2": she should name "large café capacity", state the optimistic attitude, note the -2 thousand USD downside, and explain the assumptions behind the payoffs.

Leo and Sara must do the same zoom back in. What does the selected crop imply for the farm? What risk does the project expose the investor to? Are the states complete and the monetary estimates reasonable? A model is only as useful as its alternatives, states, payoffs, units, and assumptions. A criterion is a transparent decision lens, not an automatic declaration of the one universally correct choice. Other criteria exist, but they are outside our C1–C4 journey.

1.9 A Reusable Checklist

1. What decision is being made?
2. What are the alternatives A_i?
3. What uncertain event occurs after the decision?
4. What are the states S_j?
5. What consequence belongs to each (A_i, S_j)?
6. What are the units?
7. Are payoffs supplied, or must revenue and cost be calculated?
8. Is every table cell complete?
9. Are Maximax or Maximin appropriate for the stated attitude?
10. What does the result mean in the original context?
11. What risks, assumptions, ties, or limitations must be reported?

1.10 Final Takeaway

Keep the skeleton; restore the meaning

The context gives the decision meaning. Abstraction gives it structure. The payoff table gives it visibility. A decision criterion gives it a transparent rule for comparison. Interpretation brings the mathematics back to the real world.

The stories will keep changing. Your super tool can travel with you.